

Multiple Choice (10 marks)

1. A digital signature needs a
 - (a) Private-key system
 - (b) Shared-key system
 - (c) Public-key system
 - (d) All of them
2. Public key encryption is advantageous over Symmetric key Cryptography because of
 - (a) Speed
 - (b) Space
 - (c) Key exchange
 - (d) Key length
3. _____ is a popular tool used for network analysis in multiprotocol diverse network.
 - (a) Snort
 - (b) SuperScan
 - (c) Burp Suit
 - (d) EtterPeak
4. What are the port states determined by Nmap?
 - (a) Active, inactive, standby
 - (b) Open, half-open, closed
 - (c) Open, filtered, unfiltered
 - (d) Active, closed, unused
5. A proxy rewall lters at the
 - (a) Physical layer
 - (b) Application layer
 - (c) Data link layer
 - (d) Network layer

True / False (10 marks)

Answer True or False

6. True or False: John the Ripper is primarily used for network intrusion detection and real-time traffic analysis in cybersecurity.
7. True or False: Escaping queries is an effective technique to prevent SQL injection attacks on databases.
8. True or False: The Data Encryption Standard (DES) utilizes a subkey length of 56 bits at each round of encryption.
9. True or False: The Heartbleed bug allows an attacker to read data beyond the bounds of a buffer in memory.
10. True or False: A Base Signal Station functions similarly to a Base Transceiver Station, providing signal coverage for mobile networks.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of authentication in cybersecurity and analyze how it serves to determine user identity within secure systems.
12. Discuss the concept of buffer overrun vulnerabilities in software applications. Analyze how these coding mistakes can be exploited by attackers and the implications for system security.

End of Paper